

Actionability Algebra Findings

Synthetic policy results across `actionability_algebra_v1`, `v2`, and `v3`. Focus: whether the model keeps market preference early and computes permission or actionability later in a wording-robust way.

DATE	PHASES	PROMPTS	HELD-OUT GROUPS
21 March 2026	3 synthetic runs	96 per phase	24 scenario groups

MAIN READ

The robust result is early market preference. The clean late permission result is not yet robust.

`v2` supported an “early preference, late permission” story after removing an explicit mode leak. But `v3` kept the latent rules fixed while paraphrasing and reordering section text, and the previously perfect late permission signal collapsed. The surviving claim is narrower: market-best asset is stable and early; downstream permission decoding is still fragile.

MARKET-BEST ASSET	V2 PERMISSION MODE	V3 PERMISSION MODE	TOP-SYMBOL INVARIANCE
1.000	1.000	0.625	1.000
AUROC, all phases	<code>constraints_eos L27</code>	<code>active_strategies_eos L2</code>	across permission groups

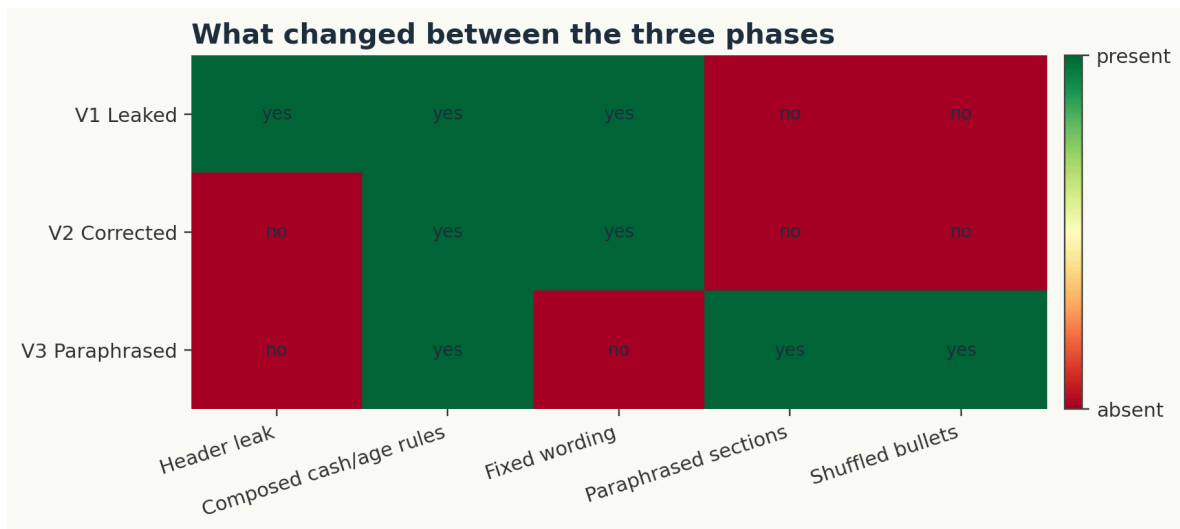
Scope

This report compares three synthetic policy phases:

- `v1`: compositional permission dataset with a prompt-header shortcut that leaked the exact `permission_mode`.
- `v2`: removed the shortcut, forcing the model to rely on the actual section content.
- `v3`: kept the same latent permission rules as `v2`, but paraphrased portfolio and constraint language and shuffled bullet order inside the sections.

The point of the sequence is not to maximize an easy classifier. The point is to see which claims survive progressively less convenient prompt wording.

Experimental Design

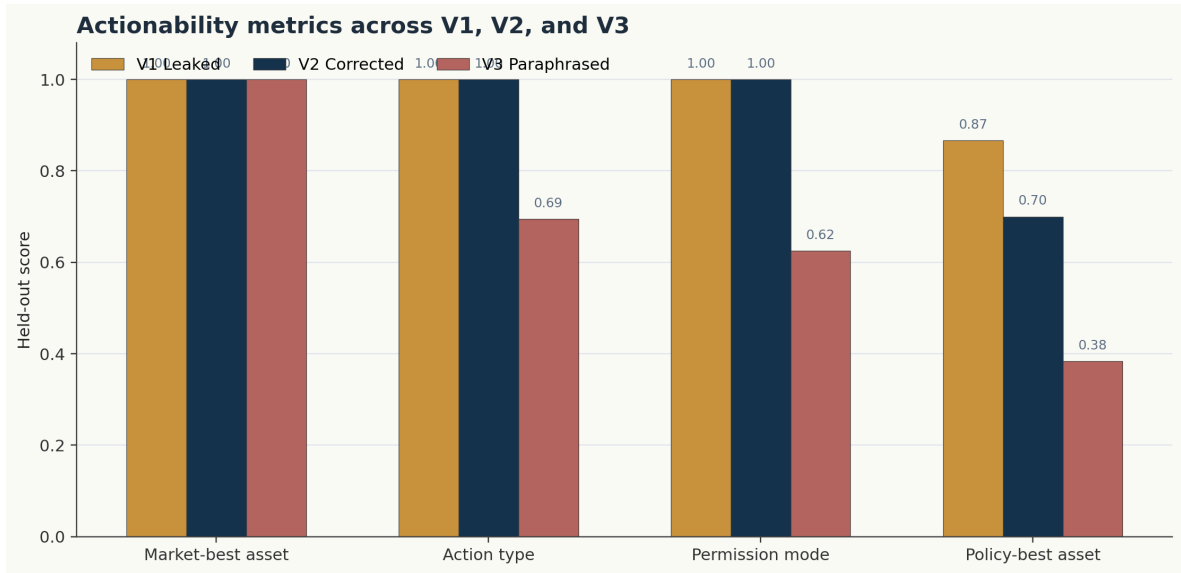


Phase design summary. v1 is useful only as a leakage baseline. v2 is the first clean read. v3 is the first robustness check against wording and local ordering changes.

Main Quantitative Findings

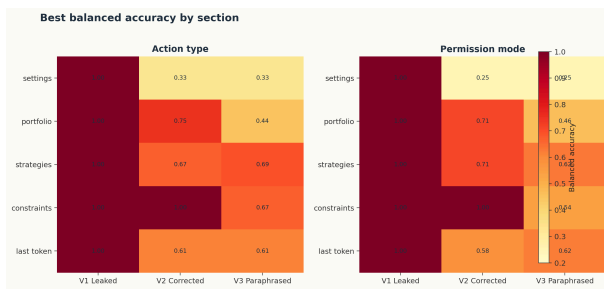
Question	v1 Best	v2 Best	v3 Best
market_best_asset	row_mean LO, AUROC 1.000	row_mean LO, AUROC 1.000	row_mean LO, AUROC 1.000
expected_action_type	active_settings_eos LO, bal acc 1.000	constraints_eos L25, bal acc 1.000	active_strategies_eos L3, bal acc 0.694
permission_mode	active_settings_eos LO, bal acc 1.000	constraints_eos L27, bal acc 1.000	active_strategies_eos L2, bal acc 0.625
policy_best_asset	portfolio_eos L5, bal acc 0.867	portfolio_eos L4, bal acc 0.700	active_strategies_eos L12, bal acc 0.383
permission_top_symbols	1.000 invariance	1.000	1.000

The critical comparison is v2 -> v3. If the latent permission signal were strongly section-semantic and wording-robust, modest paraphrase and bullet shuffling should have left the downstream read mostly intact. It did not.

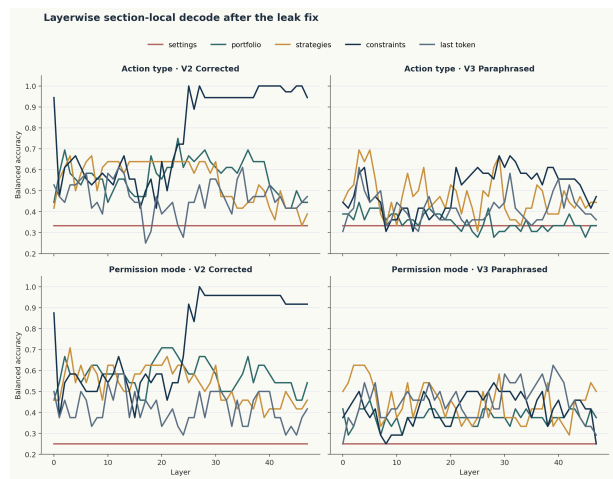


Summary metric shift across phases. Early market preference stays pinned at ceiling. The permission-side classifiers move from perfect in v2 to moderate or weak in v3.

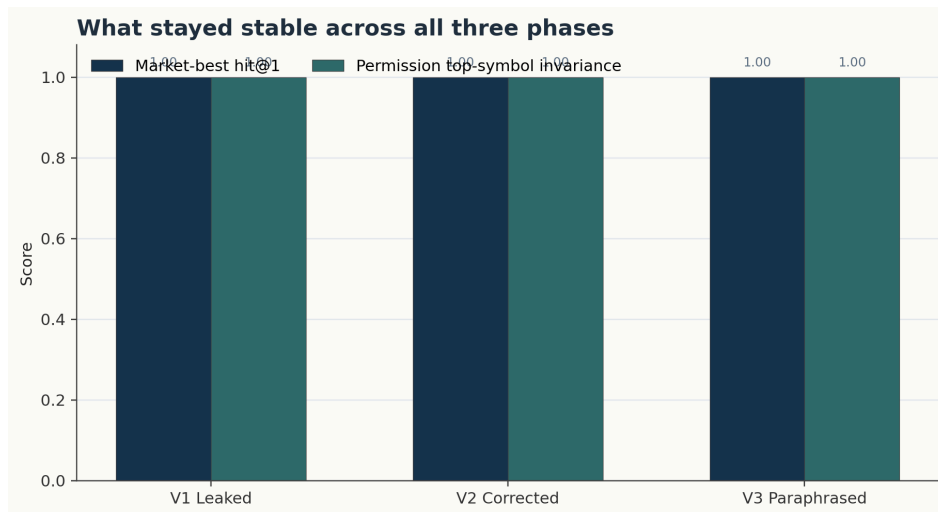
Where The Signal Lives



Best-layer heatmaps by section. v2 concentrates the permission result late in constraints_eos; v3 no longer does.



Layerwise comparison for the permission-side tasks. The v2 curve is sharp and late. The v3 curve is weaker and flatter.



Invariance stayed perfect across all three phases. The preferred top symbol under permission compositions did not move even when the permission-side classifiers weakened.

Interpretation

SUPPORTED NOW

The model's market-best asset signal is early, clean, and robust across all three prompt variants.

NOT SUPPORTED YET

A crisp, wording-robust late permission circuit is not established by the current synthetic sequence.

Best current read:

- v1 should be treated as a leakage baseline, not an interpretable result.
- v2 remains meaningful because removing the header leak moved actionability from a trivial early shortcut to a late section-local read.
- v3 is the decisive robustness check. It shows that the v2 permission result depended materially on surface presentation.
- The stable claim that survives every phase is narrower and cleaner: the model forms a market preference very early, while permission-related decoding is weaker and more presentation-sensitive.

What This Means For Direction

This result is useful because it trims the research tree:

- It is worth keeping the “early preference” line as a main hypothesis.
- It is not worth over-claiming a clean downstream permission representation from the current synthetic setup.

- The right next synthetic move is not more of v2. It is a harder v4 with paraphrases, distractor numerics, and less lexical shortcutting.
- The right real-data bridge is also narrower: validate early preference first, then treat permission as a noisy downstream gating signal.

Immediate Next Step

actionability_algebra_v4 should do four things simultaneously:

- keep the v3 paraphrase and bullet-order variation
- add distractor numeric lines inside portfolio and constraint sections
- enlarge the scenario-group set so each held-out split is less templated
- vary local wording around permission cues without changing the underlying rules

If a downstream permission read survives that stress test, it becomes a serious result. If it fails again, the better research target is likely not “permission manifold” but “preference vs permission algebra” with noisier, more distributed downstream evidence.

Actionability Algebra Findings — 21 March 2026. Three synthetic policy phases, 96 prompts per phase, 48 layers, held-out scenario-group evaluation.